

Module 11: GNN Physical Dependency Discovery and Coupling Optimization

Physics-Note Scaffold for Kinetic_MHD_PINN

Module boundary, mathematical anchor, interfaces, verification obligations, and surrogate role

Purpose of this scaffold. Infer directed physical-variable dependencies, feedback groups, state-dependent coupling strengths, and candidate numerical exchange sequences from synthetic interventions. This document establishes the module boundary before a full derivation or implementation is written. Statements labeled as planned work are not claims of a completed solver.

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Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
AE	Alfvén eigenmode	A discrete shear-Alfvén mode that can interact resonantly with energetic particles.
EP	Energetic particle	Fusion-born alpha particle, neutral-beam ion, or other suprathermal population.
MHD	Magnetohydrodynamics	Fluid description of the bulk plasma and magnetic field.
PINN	Physics-informed neural network	Neural approximation trained with governing-equation and boundary-condition residuals.
ψ	Flux-surface coordinate	Reduced radial coordinate labeling nested magnetic surfaces when such surfaces exist.
f_f	Energetic-particle distribution	Reduced or full phase-space distribution of the fast-ion population.
ω	Mode frequency	Real oscillation frequency unless a complex convention is stated.
γ	Growth rate	Positive for instability and negative for damping under the adopted convention.
U	Coupled state	Collection of fields and reduced variables exchanged among modules.

1 Scope

Module purpose. Infer directed physical-variable dependencies, feedback groups, state-dependent coupling strengths, and candidate numerical exchange sequences from synthetic interventions.

Upstream dependency. Module 8 trajectories and controlled synthetic interventions.

Primary input contract. multimodule trajectories, intervention labels, mechanism switches, time lags, and candidate graph priors.

Primary output contract. directed typed graphs, edge strengths and lags, strongly connected feedback groups, and ranked coupling strategies.

2 Mathematical starting point

The first governing-equation anchor for this module is

$$\hat{G} = \arg \min_G (\mathcal{L}_{\text{trajectory}} + \lambda_{\text{phys}} \mathcal{L}_{\text{admissibility}} + \lambda_{\text{sparse}} \|G\|_1). \quad (2.1)$$

A full version of this note must define every normalization, sign convention, coordinate system, boundary condition, initial condition, closure, and approximation used to turn this anchor into a solvable model.

3 Planned fidelity progression

1. Analytic or manufactured limit with a known qualitative or quantitative solution.
2. Reduced executable model with explicit assumptions and independent verification.
3. Coupled interface test using synthetic upstream data.
4. Higher-fidelity or external-code comparison when a trusted reference becomes available.

4 Numerical-method obligations

The implementation note for this module must specify:

- state representation and units;
- spatial, temporal, velocity-space, or spectral discretization as applicable;
- solver and convergence criteria;
- conservation, positivity, symmetry, and invariant checks;
- input/output schema and interpolation/remapping policy;
- uncertainty sources and failure flags.

5 Physics-inspired surrogate opportunity

Graph learning complements rather than replaces physical reasoning: causality, conservation, and timescale constraints remain hard admissibility filters.

A surrogate is acceptable only if its validation includes out-of-distribution tests, physical-residual checks, conserved-quantity error, and comparison against a conventional solver or trusted reference. A low data error alone is insufficient.

6 Coupling role

This module exchanges data through typed project state products rather than direct access to another solver's internal arrays. Coupling code belongs in Module 8. Any feedback edge introduced here must identify its physical mechanism, direction, natural timescale, units, and whether the exchange is explicit, lagged, or iterated.

7 Verification checklist

1. Confirm dimensional consistency and adopted normalization.
2. Recover at least one analytic or manufactured limit.
3. Demonstrate convergence with resolution or basis size.
4. Quantify sensitivity to boundary/initial conditions and closure parameters.
5. Verify relevant particle, energy, momentum, or phase-space budgets.
6. Record known nonphysical regimes and solver failure behavior.

8 Planned references

The completed note should cite primary literature for the governing model, a trusted numerical-method reference, and benchmark or validation sources. References will be selected when this module becomes an active implementation target.